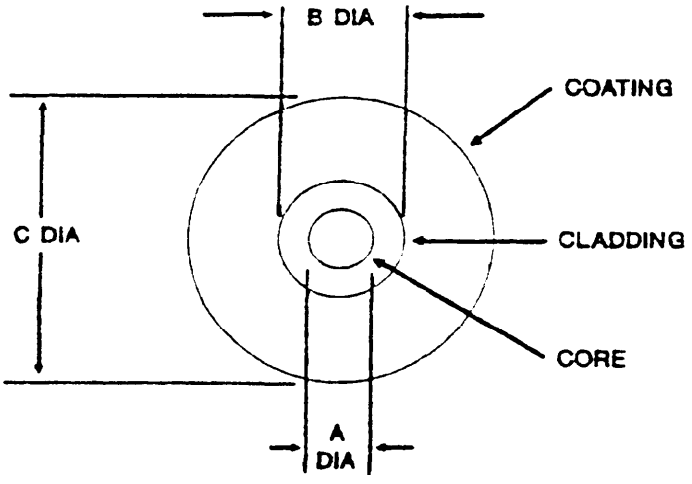
| METRIC |
MIL-F-49291/4A
8 July 1992
SUPERSEDING
DOD-F-49291/4
30 MARCH 1987

MILITARY SPECIFICATION SHEET

FIBER, OPTICAL, 100/140 MICROMETERS, (METRIC)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the optical fibers described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-F-49291



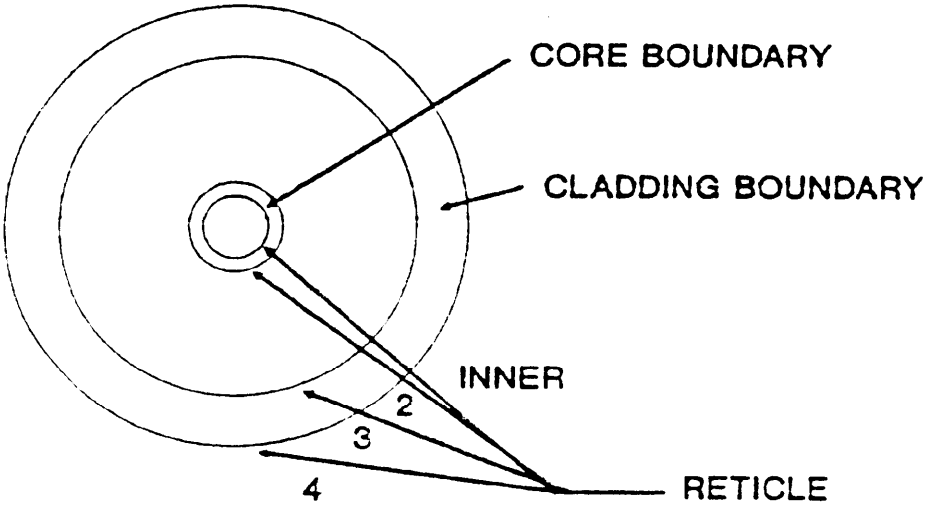
Fiber index	PIN	Dimensions		
		A	B	C
Graded	M49291/4-01	100 ± 4	140 ± 6	250 ± 15
Quasi graded	M49291/4-02	100 ± 4	140 ± 6	250 ± 15

NOTE: Dimensions are in micrometers.

FIGURE 1. Dimensions and configuration of optical fiber construction.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Circle	Diameter of circle
Inner	96 Micrometers
Second	104 Micrometers
Third	134 Micrometers
Fourth	146 Micrometers

NOTE: Dimensions are in micrometers.

FIGURE 2. Tolerance fields.

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REQUIREMENTS:

Classification:
Type: I
Class: 1 or 2 (see part number)
Composition: A
Size: V
Wavelength: A

Dimensions and configuration:
Diameter: See Figures 1 and 2.
Ovality:
Core: 10 percent
Cladding: 4 percent
Offset:
Core-to-cladding: ≤6 micrometers
Fiber-to-coating: ≤17.5 micrometers (OCCR ≥ 0.56)
Maximum percent of coating diameter change at the splice points: Not applicable.
Continuous lengths: 1100 meters minimum.

Splices: Not allowed.

Tensile strength (proof test): 690 MPa.

Fiber mass/unit length (kg/km): 0.1 kg/km maximum.

Attenuation rate: Measurements made at 850 nm ± 20 nm.

-46°C	+25°C	+71°C
3.50dB/km	3.50dB/km	3.50dB/km

Numerical aperture: 0.285 ± 0.025 at 850 nanometers ± 25 nm.

Bandwidth:
Graded index:
200 MHz·km (850nm ± 20nm)
Quasi-graded:
100 MHz·km (850nm ± 20nm)

Temperature range:
Operating: -46°C to +71°C.
Storage: -57°C to +85°C.

Fungus test: Applicable.

Nuclear radiation resistance: Not applicable.

Mechanical strippability: Applicable, except the requirements of EIA/TIA-455-178 shall not apply.

Attenuation uniformity: Not Applicable.

Macrobend attenuation: Not Applicable.

Dispersion: Not Applicable.

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PIN (see Figure 1 and table I):
M49291/4-01
M49291/4-02

TABLE I. Supersession data.

PIN	Superseding
M49291/4-01	D49291/4-01
M49291/4-02	D49291/4-02

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - SH
Air Force - 85
NASA - NA

Review activities:
Army - MI, SC
Navy - AS
DLA - ES

User activity:
Air Force - 17

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6010-0036-4)

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